

An Adaptive Signal Recovery Framework Based on Low-Rank Matrix Completion for High-Density sEMG

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Abstract—High-density surface electromyography (HD-sEMG) signals play a crucial role in a wide range of biomedical and healthcare applications by providing rich spatial electrophysiological information. However, the detection of HD-sEMG, particularly when using large-area, multichannel electronic skins, is susceptible to various noise interference caused by poor adhesion or channel failure of the electrode array, motion interference, and hardware instabilities, leading to complex and widespread signal corruption. Current signal processing and restoration methods often focus on temporal features within individual channels, without further exploiting the underlying low-rank structure and cross-channel spatial correlations inherent in high-density electrode arrays, thereby limiting the signal recovery and denoising performance in complex, noisy environments. To address these challenges, we propose an adaptive low-rank completion framework (ALF-SVT), based on singular value thresholding (SVT), for robust HD-sEMG signal recovery. The method operates directly on 2-D electromyography (EMG) maps and adaptively exploits their low-rank structure, temporal stability, and spatial correlation for signal recovery. Experimental results show that compared to conventional recovery methods, the proposed approach delivers a 64.8% improvement in average restoration accuracy, achieving an overall recovery accuracy of 86.3% under 49% signal loss or corruption. In addition,

ALF-SVT demonstrated a classification performance of 95.5% for seven hand motion classes under 40.2% signal loss or corruption. This work demonstrated that the proposed method can robustly recover signals not only in noise interference scenarios but also when they are severely distorted by large-scale data loss or hardware-induced amplitude corruption.

Index Terms—Biomedical signal processing, high-density surface electromyography (HD-sEMG), matrix completion, multichannel electrode arrays, signal recovery.

I. INTRODUCTION

THE electromyography (EMG) is an electrical signal generated by muscle fibers during neural excitation, serving as a direct representation of muscle activity and movement intent. It plays a vital role across diverse fields, such as neural prostheses control, motion monitoring, medical rehabilitation, and disease diagnosis [1], [2], [3], [4]. Traditional EMG systems typically employ sparsely and discretely distributed electrodes, resulting in poor spatial resolution [5], [6]. Recently, high-density surface EMG (HD-sEMG) signals, recorded using multichannel electrodes [7], [8], [9], such as the flexible electronic skin [10], [11], [12], shown in Fig. 1(a), have emerged as a prominent focus due to their ability to capture muscle activity with high spatial resolution. This improvement in spatial resolution enables the signals to be naturally represented in a 2-D spatiotemporal form, providing richer information for subsequent analysis and applications [13], [14].

However, as a weak bioelectrical signal, EMG is inherently susceptible to various external interferences [15], such as power line noise, motion artifacts, and electrocardiogram interference. Furthermore, HD-sEMG systems (e.g., electronic skin) face additional challenges in practical applications [16], [17]. Specifically, channel disconnections or localized sensor degradation can lead to large-scale signal loss or severe corruption. In addition, their monolithic patch-style configuration is prone to mechanical deformation (e.g., stretching or slipping) during movement, which can induce synchronous artifacts across multiple channels. These factors would greatly degrade signal quality and make the filtering and restoration of HD-sEMG data substantially more difficult.

In general, the noise interferences and data corruptions, which are hereafter referred to as signal faults, can be categorized into three typical types [see Fig. 1(b)] [18].

- 1) *Random Faults*: Isolated data points that are randomly corrupted, often due to sudden electromagnetic interferences (EMI spikes) or similar transient disturbances.

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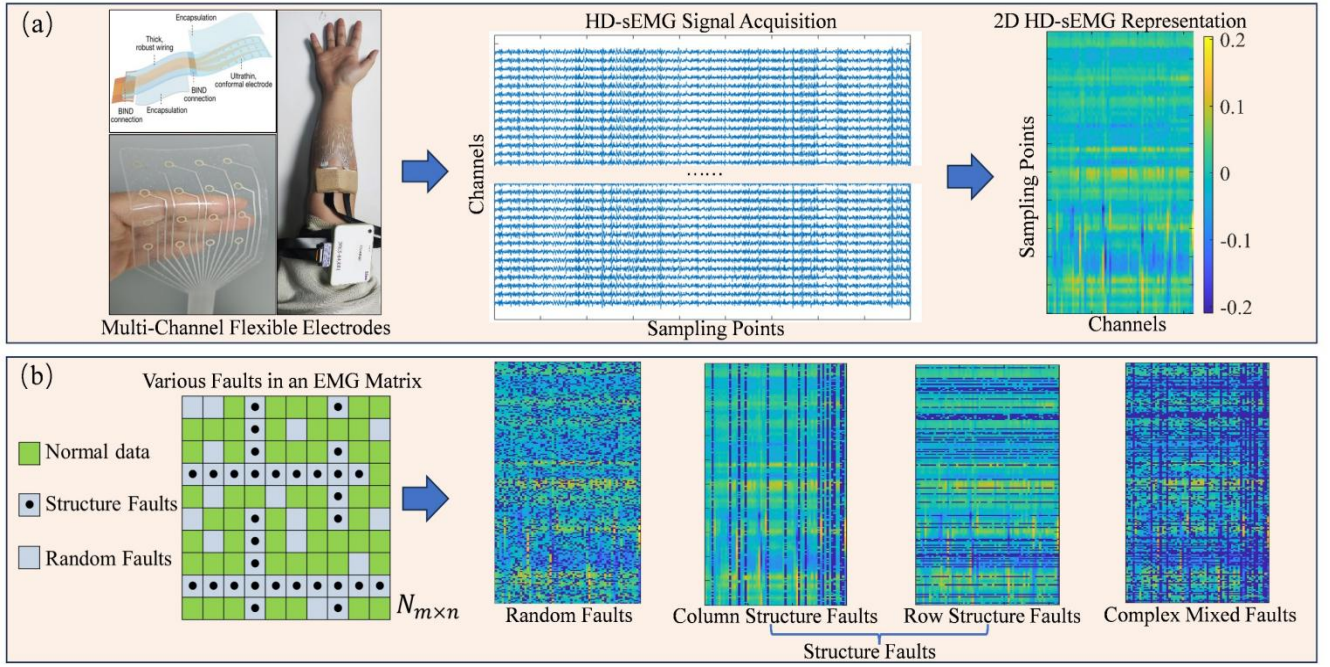


Fig. 1. (a) Two-dimensional electromyogram acquisition process from multichannel flexible electrodes. (b) Three typical types of HD-sEMG signal faults: random faults, structure faults, and complex mixed faults.

- 2) *Structure Faults*: Exhibit clear patterns and are often caused by synchronization interference or channel failures, leading to large-scale data loss along specific dimensions. Column structure faults occur when a hardware channel experiences partial failure or severe degradation, resulting in corruption of most of the data in that channel. Row structure faults arise when a specific time frame is severely corrupted, often due to transient fluctuations in the transmission bus, which result in simultaneous corruption across all channels during that period.
- 3) *Complex Mixed Faults*: Combine both random and structure faults, often manifesting as consecutive data loss along rows or columns, and interspersed with random faults, forming complex and extensive damaged regions.

To address the different types of signal faults, researchers have developed various signal processing techniques. Common approaches include bandpass filtering, power line interference suppression, and interpolation algorithms, which are used to remove noise within specific frequency ranges or fill in missing signal data [19], [20]. Independent component analysis (ICA) and low-rank matrix decomposition have also achieved some success in denoising and restoring single-channel signals [21], [22]. Recently, with the advancement of multichannel flexible electrodes, several processing methods for HD-sEMG signals have been proposed. For instance, a novel neural network-based approach, TrustEMG-Net [23], has been proposed for surface EMG signal denoising, showing promising performance. Other methods include a subspace projection-multiscale subspace (SSP-MSS) framework for suppressing multiple noise types [24], and a variational mode

decomposition (VMD)-based approach that leverages noise characteristics for targeted removal [25].

However, the above-mentioned methods designed for HD-sEMG often process each channel independently to reduce noise, rather than leveraging the inherent 2-D spatiotemporal matrix structure of HD-sEMG signals, and typically rely on specific prior assumptions about the corruption patterns or electrode configurations. As a result, their effectiveness is often limited when dealing with large-scale HD-sEMG signal loss or severe corruption under complex mixed fault scenarios, due to their inability to exploit the spatial correlation across multiple channels [26]. Therefore, to enable reliable downstream application of EMG signals, an adaptive systematic signal restoration method combining spatiotemporal information and adapting to complex fault scenarios is required to improve robust HD-sEMG signals recovery.

The contributions of this study are as follows.

- 1) We propose an adaptive low-rank completion framework (ALF_SVT) based on singular value thresholding, which operates directly on the 2-D spatial domain of HD-sEMG signals and leverages their spatiotemporal structure for recovery. The ALF-SVT achieves robust performance across diverse fault scenarios, including large-scale missing segments and severe amplitude distortions.
- 2) The framework was simulated and validated under complex mixed fault scenarios, and the signal reconstruction error rate was introduced as an additional metric alongside classification accuracy to precisely evaluate the HD-sEMG signals recovery quality.

Compared to traditional methods, the proposed ALF_SVT approach emerges as an effective and adaptive denoising

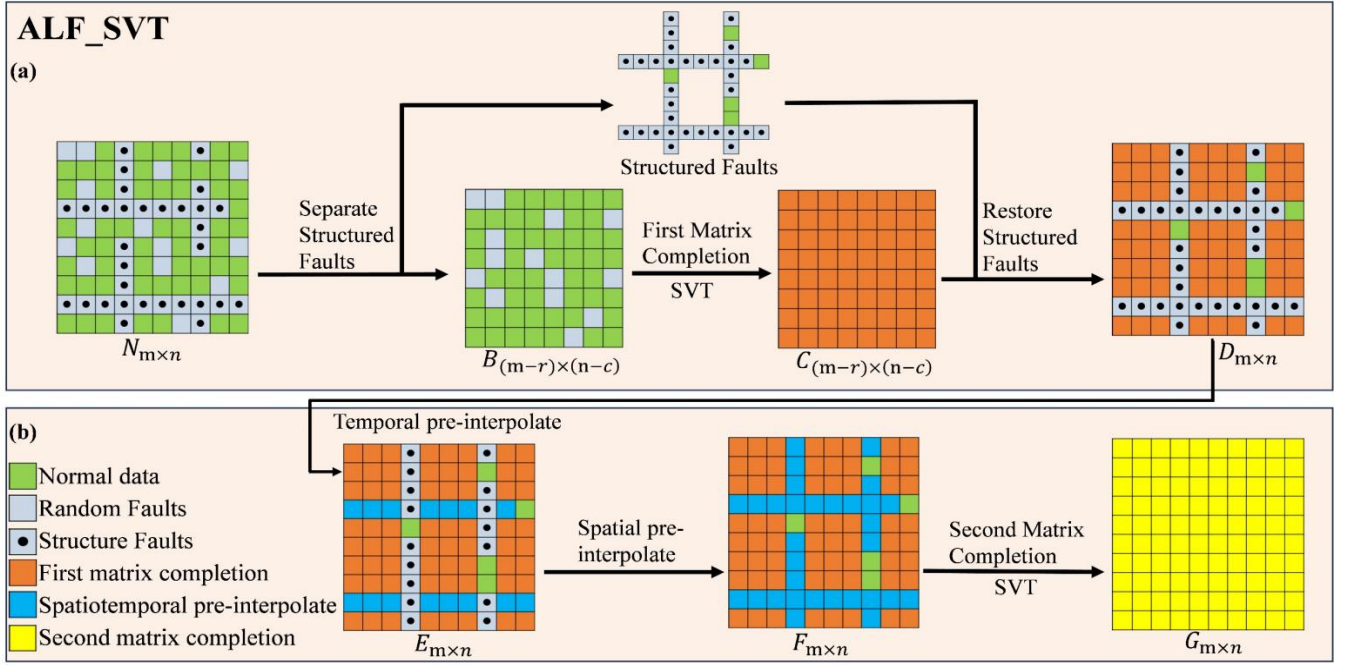


Fig. 2. ALF_SVT-based recovery process. (a) Random faults recovery. (b) Structure faults recovery.

technique that not only suppresses external interference and enhances signal quality but also maintains robust performance across varying levels and patterns of signal corruption, thereby providing a reliable dataset for downstream EMG signal applications.

II. METHODOLOGY

A. Proposed Matrix Completion-Based Adaptive Signal Restoration (ALF_SVT) Method

The problem of spatiotemporal EMG signal restoration can be formulated as a matrix completion problem, aiming to reconstruct missing or corrupted values in a structured manner. SVT assumes a low-rank structure and solves the matrix completion problem via nuclear norm minimization, as shown in the following equation, using iterative singular value shrinkage to restore corrupted data while ensuring consistency with observed entries [27]:

$$\min_X \|X\|_* \text{ s.t. } P_\Omega(X) = P_\Omega(M_{\text{obs}}) \quad (1)$$

where $\|X\|_*$ denotes the nuclear norm of matrix X , defined as the sum of its singular values, serving as a convex surrogate for the original rank minimization problem. $P_\Omega(X)$ is the projection operator that preserves the entries of X at the observed indices Ω , while zeroing out the rest. M_{obs} represents the observed data matrix.

However, SVT exhibits notable limitations under conditions of extreme sparsity or severe outliers, where it may over-compress incomplete rows or columns and fail to distinguish noise from true signals, ultimately degrading reconstruction accuracy [28].

To address the diverse types of faults in HD-sEMG signals, we propose the ALF_SVT framework, which systematically handles random faults and structure faults, as shown in

Fig. 2. The framework is composed of three main components: 1) random faults recovery; 2) structure faults recovery; and 3) spatiotemporal preinterpolation strategy.

1) Random Signal Faults Recovery: To enable structured processing, the data is segmented into windowed matrices, resulting in a matrix $N_{m \times n}$, where the number of rows m corresponds to the predefined window length in time frames, and the number of columns n corresponds to the total number of channels (in this study, $m = 200$ and $n = 64$). A half-window step size was used for sliding restoration to balance computational efficiency and temporal continuity.

a) Separate structure faults: First, to mitigate the negative impact of structure faults, ALF_SVT preprocesses the matrix by temporarily removing the rows and columns containing structure faults from the observation matrix. Rows or columns with more than 20% corrupted entries are treated as structural faults to maintain the validity of the low-rank assumption. Suppose the number of faulty rows and columns is r and c , respectively. The remaining data matrix, denoted as $B_{(m-r) \times (n-c)}$, contains only random faults.

b) First matrix completion (SVT): The matrix completion technique (SVT) is then applied to fill in the missing or damaged random faults, resulting in an initially repaired matrix $C_{(m-r) \times (n-c)}$.

c) Restore structure faults: The rows and columns with structure faults are then restored to their original positions, reverting the matrix to its original size $D_{m \times n}$.

2) Structure Signal Faults Recovery:

a) Temporal preinterpolate: Leveraging the temporal stability of EMG signals, the rows affected by structure faults are prefilled using time data from neighboring sensors, resulting in a partially restored matrix $E_{m \times n}$.

b) *Spatial preinterpolate*: Similarly, based on spatial correlation and the principle of low-ranking, the columns with structure faults are prefilled spatially, producing the matrix $F_{m \times n}$.

c) *Second matrix completion (SVT)*: In the end, the SVT matrix completion technique is applied to the processed matrix. By combining low-rank characteristics and spatiotemporal stability, the original data matrix $G_{m \times n}$ is restored.

Based on Cai et al. [27] and preliminary experiments, the singular value threshold was set as $\tau = 2(n_1 \times n_2)^{1/2}$. The iteration was terminated when the relative change between two consecutive updates fell below $\varepsilon = 10^{-4}$ to ensure convergence efficiency and reconstruction accuracy.

3) *Spatiotemporal preinterpolation strategy*:

a) *Temporal Preinterpolation*: The exponentially weighted moving average (EWMA) is an effective estimation technique commonly used in time series modeling [29]. Based on EWMA, we designed a temporal preinterpolation algorithm that uses the four nearest nonfaulty time points (i.e., the two closest valid time points before and after the fault) to prefill the data missing due to column structure faults. The specific implementation is presented in the following equation:

$$M_i = \lambda \left(\frac{M_{i+1} + M_{i-1}}{2} \right) + (1 - \lambda) \left(\frac{M_{i+2} + M_{i-2}}{2} \right). \quad (2)$$

Here, λ is the smoothing constant, while M_{i+1} , M_{i-1} , M_{i+2} , and M_{i-2} represent the nearest and second-nearest valid frames before and after the faulty point M_i . In this study, the smoothing constant λ is set to 0.8 to maintain stable prefilling performance.

b) *Spatial Preinterpolation*: Based on spatial correlation, the estimation for calculating the mean of neighboring channels is formulated as shown in the following equation:

$$M_m(t, k) = \begin{cases} M(t, k), & (t, k) \notin \Omega \\ \frac{1}{|C(k)|} \sum_{j \in C(k)} M(t, j), & (t, k) \in \Omega. \end{cases} \quad (3)$$

In this context, $M(t, k)$ represents a point on the 2-D EMG map, Ω is the set of erroneous data, and $C(k)$ denotes the set of correlated channels for the damaged channel k . The repair process is achieved by calculating the mean of the correlated channels. The set of correlated channels $C(k)$ is based on the arrangement of the actual valid neighboring channels and is defined as $C(k) = \{j \mid j \in \{k - b, k + b, k - 1, k + 1\}, 1 \leq j \leq n \setminus (B(k) \cup E_{\text{bad}})\}$, where the spatial electrode array arranged in an $w \times b$ grid, E_{bad} represents the set of faulty channels, and $B(k)$ denotes the set of invalid channels due to boundary constraints. The set $B(k)$ is defined as follows:

$$B(k) = \begin{cases} \{k - 1\}, & \text{if } k \bmod b = 1 \text{ (top boundary)} \\ \{k + 1\}, & \text{if } k \bmod b = 0 \text{ (bottom boundary)} \\ \{k - b\}, & \text{if } k \leq b \text{ (left boundary)} \\ \{k + b\}, & \text{if } k > n - b \text{ (right boundary)}. \end{cases} \quad (4)$$

Combining the temporal correlation of the damaged channel data, we further refine the spatial estimation by applying singular spectrum analysis (SSA) to the previously restored

columns. By retaining the dominant singular values, an estimated signal M_{SSA} is obtained and combined with M_m to enhance the accuracy of prefilling, as outlined in the following equation:

$$M_{\text{filled}}(t, k) = \begin{cases} M(t, k), & (t, k) \notin \Omega \\ \lambda \cdot M_{\text{SSA}}(t, k) \\ + (1 - \lambda) \cdot M_m(t, k), & (t, k) \in \Omega. \end{cases} \quad (5)$$

B. Signal Restoration Methods for Comparison

Previously, we discussed the matrix completion-based data recovery framework (ALF_SVT) in the context of multi-channel flexible electrodes. However, in practical applications, traditional methods such as temporal mean filtering and spatial mean interpolation remain the most commonly used approaches [30], [31], [32]. These methods are widely adopted for signal restoration tasks due to their simplicity and low computational cost. In addition, a recent deep learning-based method (TrustEMG-Net) and a classical low-rank optimization approach, augmented Lagrange multiplier (ALM), were also included.

1) *Temporal Mean Filtering Method*: The temporal mean filtering method (TM) is a classical signal restoration technique based on the time domain. It assumes local smoothness of the time-series signal, meaning that adjacent time points are statistically correlated and the signal varies gradually over time.

Let the input signal be a single-channel time series $S(t)$ with a domain of $t \in \{1, 2, \dots, T\}$. Some values of $S(t)$ are missing, forming a missing dataset $\Omega \subseteq \{1, 2, \dots, T\}$, and $S'(t)$ denotes the restored signal value. The repair of missing data is described by the following equation:

$$S'(t) = \begin{cases} S(t), & t \notin \Omega, \\ \frac{1}{|N(t)|} \sum_{k \in N(t)} S(k), & t \in \Omega \end{cases} \quad (6)$$

where $N(t) = \{k \mid k \in [t - W, t + W] \cap (\{1, 2, \dots, T\} \setminus \Omega)\}$ represents the set of observed (nonmissing) data points within a symmetric window of size $2W + 1$ centered at t . The cardinality $|N(t)|$ indicates the number of available observations around time t .

2) *Spatial Mean Interpolation Method*: The spatial mean interpolation method (SM) partially overlaps with the previously introduced spatial preinterpolation approach. It relies solely on spatial correlation by calculating the mean of neighboring channels to perform signal prefilling. The detailed formulations are provided in (3) and (4).

3) *TrustEMG-Net Denoising Method*: TrustEMG-Net [23] is a denoising autoencoder that integrates a U-Net structure with a Transformer encoder, leveraging both local feature extraction and global dependence modeling for EMG restoration. To ensure a fair comparison with the proposed ALF_SVT framework, the TrustEMG-Net model was retrained on our dataset using the corrupted-point mask within the same preprocessing pipeline.

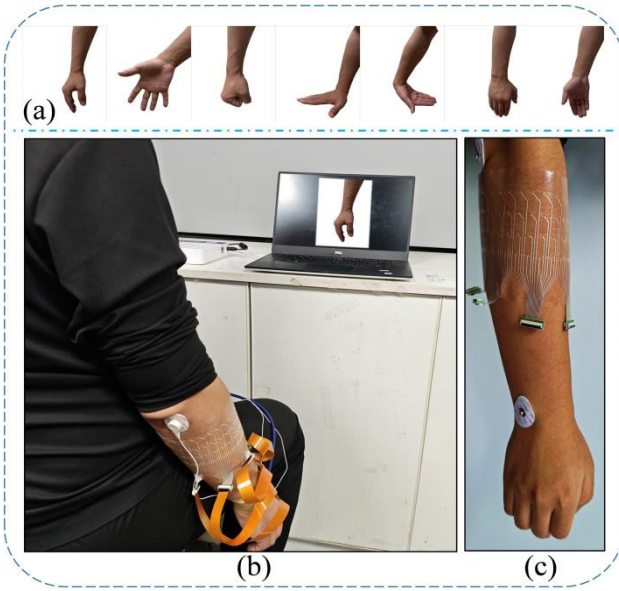


Fig. 3. HD-sEMG collection. (a) Seven hand gestures performed during data collection. (b) HD-sEMG recording setup. (c) Multichannel flexible electrode array.

4) *ALM Method*: The ALM [33] is a widely used low-rank matrix completion technique that decomposes the observed signal matrix into low-rank and sparse components through iterative Lagrange multiplier updates. It provides a stable optimization framework for structured signal restoration and serves as a representative baseline for evaluating the performance of low-rank recovery methods.

C. Data Acquisition and Preprocessing

The experimental protocol was approved by the Institutional Review Board of Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences (SIAT-IRB-221115-H0619), and all subjects provided consent for the publication of photographs for scientific and educational purposes.

In this experiment, HD-sEMG signals were collected from eight healthy subjects. Both the high-performance acquisition system (NES-64B01, SIAT, China) and the high-density, flexible, and stretchable electrode array [34] were self-developed. The acquisition system operates at a sampling rate of 2000 Hz. This electrode's flexible and stretchable design ensures good conformity to the skin surface during muscle contractions. As shown in Fig. 3(b), the electrode patch was placed on the lower third of the forearm. It contains 64 channels evenly distributed in a 16×4 grid with 1-cm spacing, as illustrated in Fig. 3(c).

During data collection, participants were instructed to perform seven distinct gestures [see Fig. 3(a)], each repeated six times and held for 4 s: no motion (NM), hand open (HO), hand close (HC), wrist extension (WE), wrist flexion (WF), wrist pronation (WP), and wrist supination (WS). To prevent muscle fatigue, a 4-s rest was given between repetitions of the same gesture, and a 30-s rest between different gestures. Then, the acquired sEMG signals were preprocessed using a 50-Hz notch filter and a 10–500-Hz bandpass filter.

D. Simulation of Various Signal Fault Models

To investigate the performance of our proposed signal restoration or ALF_SVT method, we simulated two types of signal corruption, namely, random faults and structure faults. In both cases, zero-mean Gaussian white noise is added to the selected points in the signal, with the noise intensity controlled by a specified signal-to-noise ratio (SNR) as follows.

1) *Random Faults*: $\beta\%$ of the points are randomly selected across the HD-sEMG signal matrix, and Gaussian noise is added at those positions.

2) *Structure Faults*: Specific fault rows and fault columns are selected in the HD-sEMG matrix, within which $\alpha\%$ of the data points are randomly corrupted with Gaussian noise.

3) *Complex Mixed Faults*: Combine both random and structure faults, with Gaussian noise applied to the selected points.

Totally, by adjusting the parameters of β , α , SNR, fault rows, and fault columns, we could simulate different signal fault patterns with varying corrupted spatial extent and intensity.

E. Performance Evaluation and Statistical Analysis

To assess the low-rank and spatiotemporal characteristics of HD-sEMG signals, the retained energy captured by the first k singular values of each temporal window was quantified using singular value decomposition (SVD) [35]

$$E_k = \frac{\sum_{i=1}^k \sigma_i^2}{\sum_{i=1}^{\min(T,N)} \sigma_i^2} \quad (7)$$

where σ_i is the i th singular value.

For temporal stability analysis, we computed the difference between adjacent intervals defined as $G_gap(t, k) = |X(t, k) - X(t-1, k)|/q$, where $X(t, k)$ denotes the EMG signal at time t and channel k , and q is the 99.9th percentile of the absolute difference used for normalization. For spatial correlation analysis, the Pearson correlation coefficients between adjacent channels in the 2-D electrode array, defined as $\rho_{i,j} = \text{cov}(X_i, X_j) / [\sigma_{(X_i)} \sigma_{(X_j)}]$, where X_i and X_j are the signals of channels i and j , cov denotes covariance, and σ is the standard deviation.

To evaluate the effectiveness of the restoration method, suitable metrics are needed to measure the difference between the repaired and true signals. In this study, the signal reconstruction error rate, subsequently referred to as error rate [17], quantifies the proportion of the reconstruction error relative to the true values, is used as the evaluation metric to assess the signal restoration performance, and is defined as follows:

$$\text{Error Rate} = \frac{\|M_{\text{true}} - M_{\text{est}}\|_F}{\|M_{\text{true}}\|_F} \quad (8)$$

$$\|M_{m \times n}\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n M_{ij}^2} \quad (9)$$

where $M_{\text{true}} \in R_{T \times N}$ represents the true signal matrix, $M_{\text{est}} \in R_{T \times N}$ is the repaired signal matrix, and R represents the set of matrix elements. $\|\cdot\|_F$ denotes the Frobenius norm, used to measure the overall error of the matrix. The Frobenius norm is computed only over the initially corrupted points.

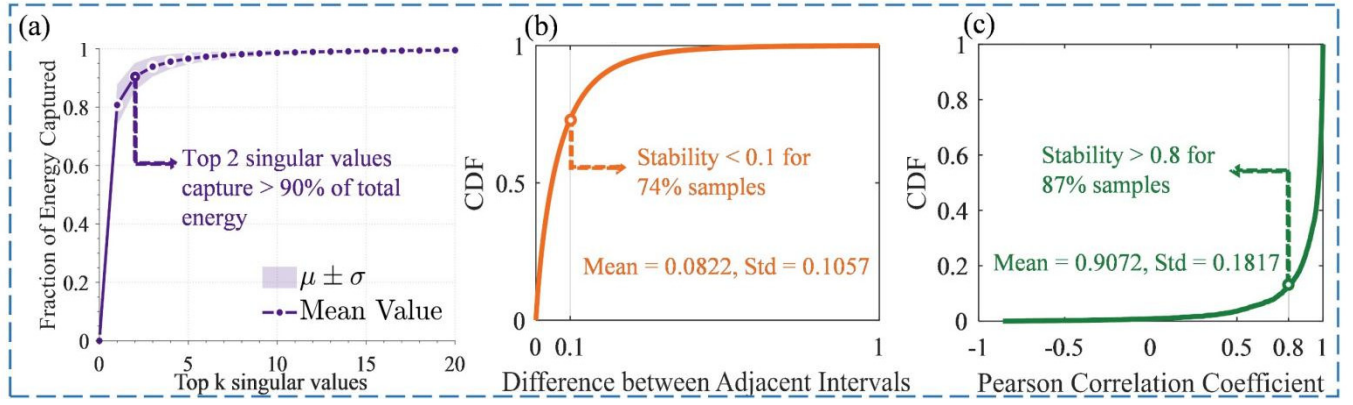


Fig. 4. Analysis of low-rank and spatiotemporal properties in HD-sEMG. (a) Mean energy distribution for the first k singular values. (b) CDF of temporal stability between adjacent time intervals. (c) CDF of spatial correlation between adjacent channels in the 2-D electrode array.

By using the error rate, the metric is independent of the absolute size of the signal matrix, making it suitable for cross-comparison between different datasets. The error rate also reflects the overall discrepancy between the repaired matrix and the actual matrix, i.e., the smaller the value, the closer the repair result is to the true value. An overall recovery rate can also be defined to quantify the total signal recovery.

To assess the functional quality of the repaired signals, four common time-domain features, namely, mean absolute value (MAV), waveform length (WL), zero crossing (ZC), and slope sign changes (SSCs), were extracted [36]. Linear discriminant analysis (LDA) was used for multiclass classification across seven actions performed by eight subjects, as shown in Fig. 3(a). The recognition accuracy was calculated as

$$\text{Recognition Accuracy} = \frac{N_{\text{correct}}}{N_{\text{total}}} \times 100\%. \quad (10)$$

Furthermore, statistical significance analysis was conducted using independent two-sample t -tests to evaluate whether the performance differences between ALF_SVT and the other methods (TM, SM, ALM, TrustEMG-Net, and SVT) were statistically significant. If the resulting p -values were less than 0.05, 0.01, or 0.001, the corresponding results were annotated with *, **, and ***, respectively.

III. RESULTS

To systematically evaluate the effectiveness of the proposed multichannel signal restoration method, we first assess the applicability of the low-rank matrix completion strategy to HD-sEMG restoration. We then perform a comparative analysis across three typical fault types, namely: random, structural, and complex mixed faults, to evaluate restoration performance under varying corruption patterns. Finally, we validate the practicality of the proposed ALF_SVT by comparing its classification performance with other restoration methods in the context of prosthetic control.

A. Analysis of Low-Rank and Spatiotemporal Properties in HD-sEMG Signals

Given that our restoration strategy is based on low-rank matrix completion, which incorporates temporal stability and

spatial correlation during the prefilling step. Therefore, to assess the applicability of low-rank matrix completion, we first analyzed the mean energy distribution captured by the first k singular values across each temporal window. As shown in Fig. 4(a), the first two singular values account for over 90% of the retained signal energy, further validating the low-rank property of the data and supporting the rationale for using low-rank approximation in signal restoration [27].

For temporal stability analysis, the cumulative distribution function (cdf) of G_{gap} , shown in Fig. 4(b), indicates strong temporal stability in HD-sEMG signals, with 74% of temporal differences below 0.1 and a mean value of 0.0822 (std = 0.1057).

For spatial correlation analysis, Fig. 4(c) presents the Pearson correlation coefficients between adjacent channels in the 2-D electrode array, with approximately 87% exceeding 0.8 and a mean value of 0.9072 (std = 0.1817), indicating strong interchannel correlations.

B. Comparative Analysis of HD-sEMG Restoration for Different Signal Faults

1) HD-sEMG Restoration Under Random Signal Faults:

To evaluate the restoration performance under random faults, we randomly corrupted $\beta = 5\%$ – 50% of the HD-sEMG signal and added Gaussian white noise at an SNR of -30 dB.

Fig. 5 illustrates the mean error rate under varying random fault corruption ratios ($\beta = 5\%$ – 50%). The mean error rates of TM (ranging from 0.21 to 0.42), SM (from 0.81 to 0.90), TrustEMG-Net (from 0.11 to 0.46), ALM (from 0.19 to 0.60), and ALF_SVT (from 0.12 to 0.21) all exhibit a relatively slow and approximately linear increase as β grows. In contrast, SVT shows a pronounced and nonlinear escalation in error rate, rising from 0.14 to 1.32 as β increases, indicating its limited robustness under higher corruption levels. Notably, ALF_SVT maintains the lowest error rates across all corruption levels, showing strong robustness to random faults.

2) HD-sEMG Restoration Under Structure Signal Faults:

To evaluate the restoration performance under structure faults, tests were conducted on each channel, with only one channel corrupted by structural noise in each test, while the others

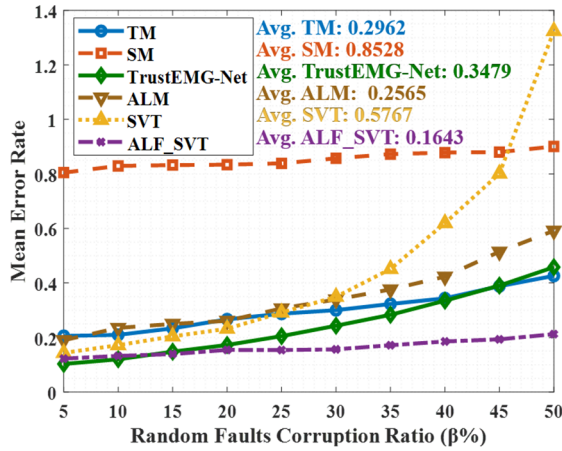


Fig. 5. Mean error rate under varying random fault corruption ratios ($\beta = 5\%$ – 50%).

remained intact. $\alpha = 10\%$ – 100% of the points in the selected channel were corrupted with Gaussian white noise.

First, Fig. 6(a) illustrates the experimental paradigm and a representative example of signal restoration, including the original signal, corrupted signal, and the restored signals obtained by TM, SM, TrustEMG-Net, ALM, SVT, and ALF_SVT. Compared to the other methods, ALF_SVT produces a restoration curve most similar to the original signal.

Second, to evaluate performance under varying structural fault conditions, we analyze the restoration error rates averaged across all 64 channels at a fixed noise level (SNR = -30 dB). As illustrated in Fig. 6(b), we compare six methods, TM, SM, ALM, TrustEMG-Net, SVT, and ALF_SVT, over a range of corruption ratios (α). For TM, the error rate increases steadily from 0.17 to 1.00 as α grows, with an average of 0.484 ± 0.110 (mean $\pm$ std). Among all methods, SM shows a flatter trend, with the error rate fluctuating between 0.71 and 0.73 as α increases. However, the standard deviation of 0.35 indicates greater variability across channels, suggesting instability in different signal conditions. ALM exhibits a wider range, increasing from 0.13 to 1.26 (mean = 0.5274), while TrustEMG shows a slightly smoother trend, varying from 0.12 to 1.23 with a mean of 0.5662. SVT maintains relatively low error rates at lower corruption levels, increasing from 0.18 to 0.63 when $\alpha < 0.7$. Beyond this threshold, the error rate rises sharply, exceeding 6.12 at $\alpha = 1.0$, resulting in a high average error of 1.341 ± 1.15 , highlighting its sensitivity to severe corruption. In comparison, ALF_SVT maintains low and stable error rates from 0.13 to 0.31 when $\alpha < 1.0$. Although the error rate rises to 0.95 under full corruption ($\alpha = 1.0$), it still significantly outperforms SVT across most corruption levels. The underlying reason for this performance drop and a mitigation strategy are further discussed in Section IV.

At last, the results in Table I present the average restoration error rates under different SNR levels (ranging from 30 to -40 dB) for each method, with the corruption ratio fixed at $\alpha = 0.8$. TM, SM, and ALF_SVT maintain relatively stable performance across the full SNR range, with average

TABLE I
AVERAGE ERROR RATES OF DIFFERENT METHODS ACROSS 64 CHANNELS

SNR(dB)	Restoration Error Rates under Varying SNR Levels					
	TM	SM	TrustEMG-Net	ALM	SVT	ALF_SVT
30	0.691	0.719	0.102	0.190	0.374	0.235
20	0.687	0.722	0.173	0.235	0.369	0.230
10	0.699	0.723	0.257	0.261	0.381	0.242
0	0.685	0.723	0.345	0.313	0.375	0.243
-10	0.672	0.721	0.487	0.452	0.384	0.239
-20	0.675	0.718	0.652	0.644	0.378	0.235
-30	0.678	0.720	0.861	0.829	1.598	0.234
-40	0.673	0.720	1.276	1.432	40.588	0.243
AVERAGE	0.683***	0.721***	0.519***	0.545***	5.556***	0.238

Significant differences compared with the proposed method:

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

error rates of 0.683, 0.721, and 0.238, respectively. TrustEMG and ALM exhibit gradual degradation with decreasing SNR, maintaining low errors under moderate noise and reaching 0.519 and 0.545 on average, respectively. In contrast, SVT performs well under mild to moderate noise (SNR = 30 to -20 dB, error rate fluctuating around 0.37), but its performance deteriorates sharply as the noise intensifies. At -30 dB, the error rate surges to 1.598, and at -40 dB, it reaches an extreme value of 40.588. This drastic degradation under severe interference is consistent with the behavior observed in Fig. 6(b).

3) *HD-sEMG Restoration Under Complex Mixed Signal Faults*: To evaluate the restoration performance under complex mixed faults, we set $\beta = 20\%$ for random faults and additionally introduced structure faults by randomly selecting different numbers of rows and columns, with $\alpha = 80\%$, and the corrupted points were subjected to Gaussian white noise at an SNR of -30 dB.

Fig. 7(a) illustrates the impact of increasing the number of fault rows and columns on the restoration error rate under mixed fault conditions. TM exhibits a steady increase in error rate, ranging from 0.35 to 0.69, with an average of 0.55. SM is largely insensitive to fault extent, maintaining stable performance but with a consistently high error rate of around 0.84. ALM shows growing instability, with the error rate increasing from 0.24 to 1.25, while TrustEMG-Net exhibits a smoother trend, rising gradually from 0.27 to 0.89. SVT experiences growing degradation, with the error rate rising sharply from 0.23 to 0.98 and averaging 0.65. In contrast, ALF_SVT achieves the lowest and most stable performance, maintaining error rates between 0.15 and 0.28 and an average of 0.23. Compared to the average of the other three methods, ALF_SVT delivers a 64.8% improvement in restoration accuracy under mixed fault conditions.

Fig. 7(b) provides a comparison of the ALF_SVT restoration result under the challenging noise configuration (16 corrupted columns, 50 corrupted rows, and 20% randomly corrupted points). From left to right, the original 2-D EMG signal, the corrupted input, and the restored output. Approximately 49% of the data points were corrupted in the corrupted

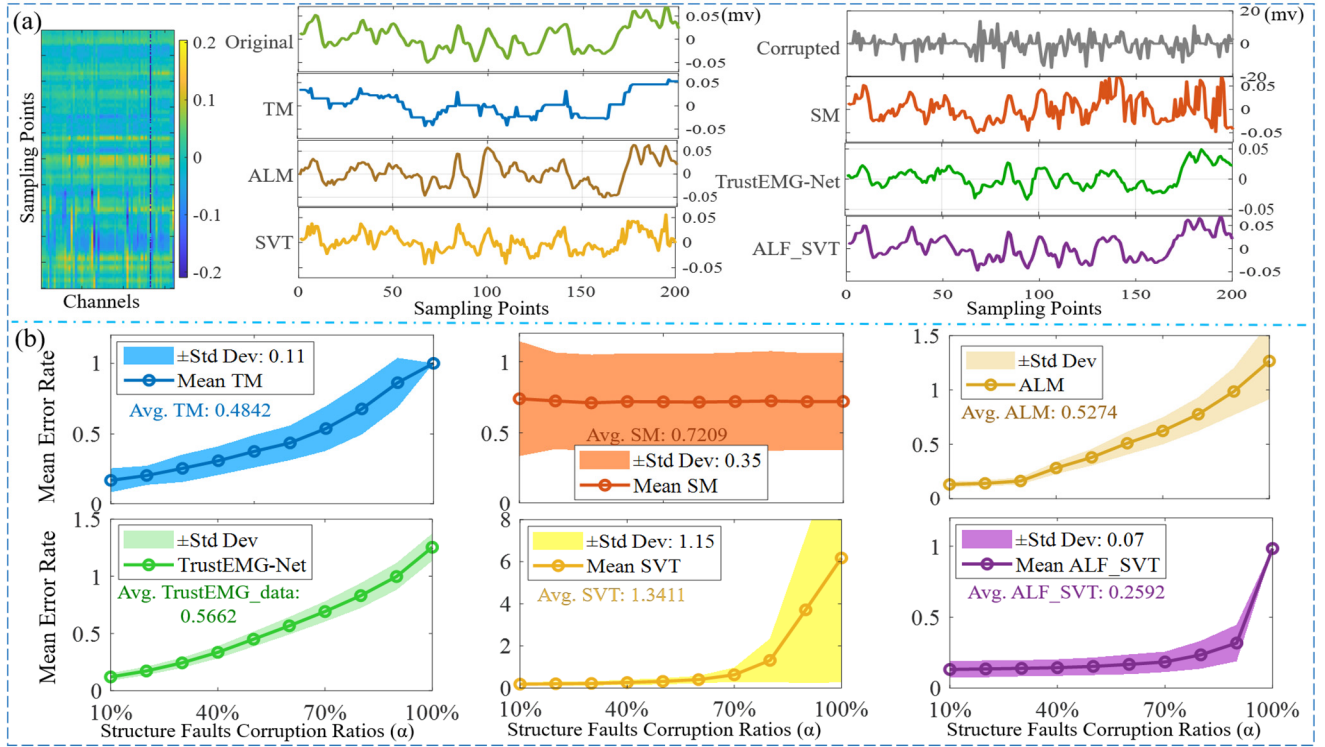


Fig. 6. (a) Paradigm and comparison of different signal restoration methods under structure signal faults. (b) Average restoration error rates across 64 channels under varying structure signal faults corruption ratios (α).

input, representing one of the most severe interference cases considered in this study. The restored output corresponds to the ALF_SVT result in Fig. 7(a), where the error rate was reduced to 0.28, and the overall recovery accuracy achieved 86.3%.

In contrast, Fig. 7(c) and (d) provides supplementary details on the restoration performance of ALF_SVT. Fig. 7(c) shows the error rate of the normal uncorrupted data after restoration, which remains on the order of 10^{-3} across all cases. This result demonstrates that ALF_SVT preserves the integrity of clean signals, introducing negligible distortion during the restoration process. Fig. 7(d) presents the repair error rate under missing data conditions, where corrupted values are replaced with zeros instead of noise. The results demonstrate that ALF_SVT maintains consistent restoration accuracy in both noisy and missing data scenarios, confirming its robustness across different types of signal degradation.

C. Multiple Gesture Recognition of the Proposed ALF_SVT Compared to Other Restoration Methods

To evaluate the effectiveness of ALF_SVT in practical pattern recognition tasks, we set $\beta = 20\%$ for random faults and introduced structure faults by selecting 15 fault columns and 20 fault rows, with 80% of their points ($\alpha = 80\%$) corrupted using Gaussian white noise at an SNR of -30 dB, resulting in 40.2% of the total signal samples being corrupted.

Fig. 8 shows the time-domain and time-frequency comparisons of signal restoration results from different methods. In the time domain, TM, SM, TrustEMG-Net, ALM, and SVT exhibit noticeable deviations from the original

waveform, whereas ALF_SVT achieves a much closer match. In the time-frequency domain, ALF_SVT also demonstrates a distribution that more closely resembles the original signal compared to the other methods. Furthermore, the performance of TM, SM, TrustEMG-Net, ALM, SVT, and ALF_SVT was evaluated on seven gestures, using an LDA classifier across all eight subjects, as shown in Table II. It can be observed that the corrupted signals yielded an average accuracy of $47.7\% \pm 15.2\%$, significantly lower than the original data, highlighting the impact of signal faults. After restoration, ALF_SVT showed significant improvements ($p < 0.01$) compared to other methods, achieving the highest recognition accuracy of $95.5\% \pm 1.5\%$, outperforming TM ($86.4\% \pm 4.4\%$), SM ($89.2\% \pm 3.0\%$), TrustEMG-Net ($90.3\% \pm 3.6\%$), ALM ($81.1\% \pm 3.9\%$), and SVT ($82.8 \pm 4.2\%$).

D. Recovery Under Realistic Interference Conditions

To evaluate the robustness of the proposed method under practical interference, two representative scenarios were tested, non-Gaussian and dynamic motion interference, as illustrated in Fig. 9. The first row shows the recovery performance under 50 Hz harmonic interference, where the SNR improved from -5 to 24.8 dB after ALF_SVT restoration. The second row presents the recovery under dynamic motion interference, simulated by a rapid wrist flexion-extension-pronation task with slight electrode pressing to induce transient contact loss and amplitude drift. The restored 2-D EMG map exhibited improved temporal smoothness and structural consistency, with the SNR increasing by approximately 8.51 dB, confirming the effectiveness of the proposed method.

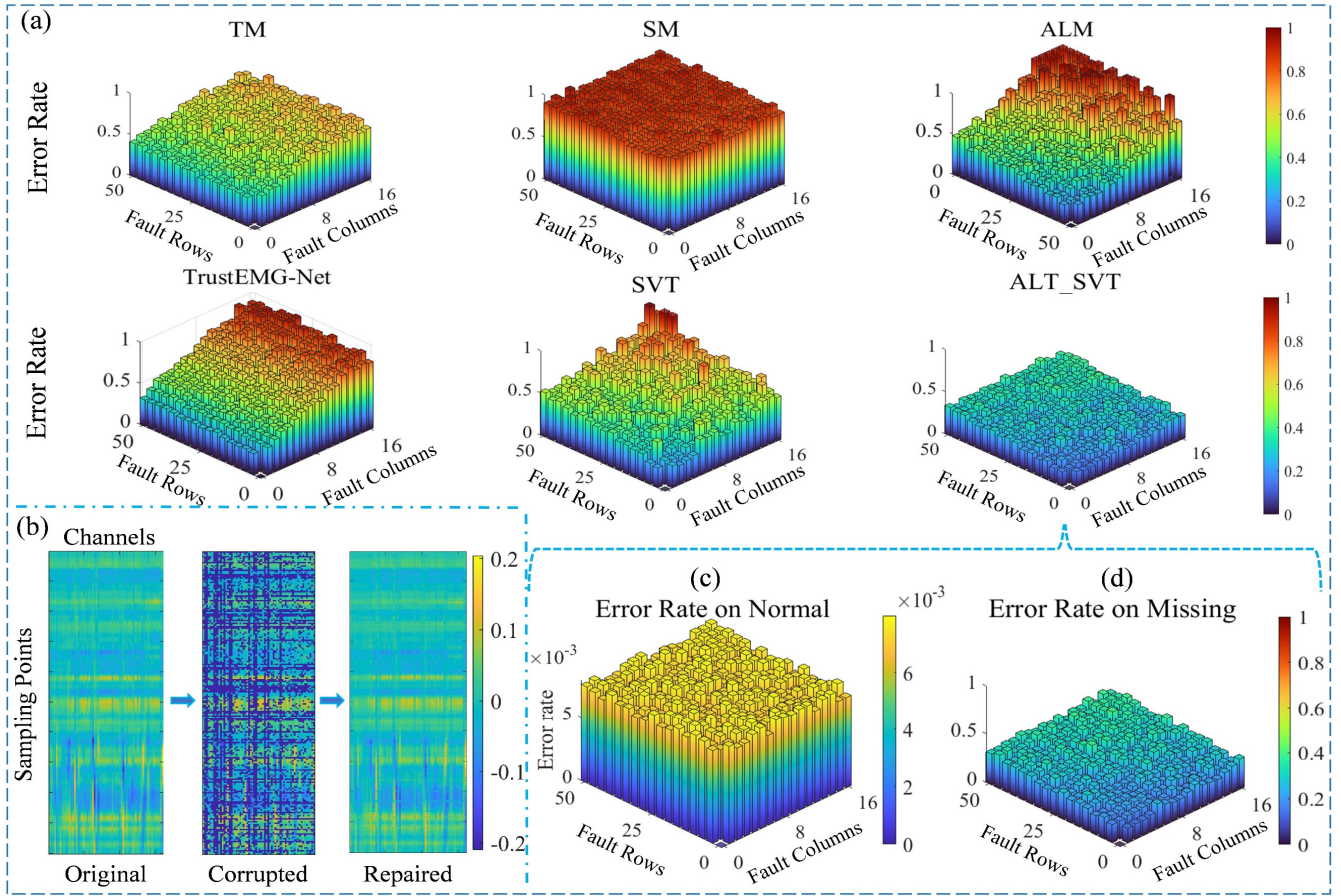


Fig. 7. (a) Repair error rates for six methods under complex mixed faults: TM, SM, ALM, TrustEMG-Net, SVT, and ALF_SVT. (b) Schematic of the 2-D EMG repair. (c) Error rate of the normal data after ALF_SVT restoration. (d) Repair error rate of ALF_SVT under the set missing conditions.

TABLE II
CLASSIFICATION ACCURACY OF THE HAND GESTURES BASED ON RESTORED SIGNALS OF DIFFERENT METHODS

MOTION	Recognition Accuracy Comparison Among Methods (%)							
	Original	Corrupted	TM	SM	TrustEMG-Net	ALM	SVT	ALF_SVT
NM	99.3	80.8	94.4	92.8	90.2	81.1	90.3	96.9
HO	98.5	40.6	90.4	91.1	97.3	89.6	84.9	97.8
HC	96.4	47.1	83.9	90.9	88.0	80.3	82.2	95.2
WE	97.1	47.6	86.3	88.2	93.0	82.4	84.8	95.5
WF	95.6	39.1	83.7	87.0	87.8	77.6	78.0	93.8
WP	95.7	36.0	82.4	84.0	85.8	77.1	79.4	94.0
WS	95.8	42.5	83.6	90.3	89.9	79.9	80.2	95.0
AVERAGE	96.9±1.5	47.7±15.2	86.4±4.4**	89.2±3.0***	90.3±3.6**	81.1±3.9***	82.8±4.2***	95.5±1.5

Significant differences compared with the proposed method: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

IV. DISCUSSION

HD-sEMG signals inherently exhibit spatiotemporal low-rank characteristics due to the physiological and spatial correlations across muscles and electrode channels [37], [38], as illustrated in Fig. 4. The results prove that the first two singular values retain over 90% of the signal energy, indicating a strong low-rank structure. Temporal stability and spatial correlation were also assessed to support the

prefilling strategy. These findings demonstrate the suitability of applying ALF_SVT to HD-sEMG signal restoration under low-rank and spatiotemporal regularities.

We compare the denoising effects of varying degrees of random faults, structural faults, and complex mixed faults on the proposed method, as depicted in Figs. 5–7 and Table I. ALF_SVT maintains a low reconstruction error (typically below 0.3) even as the SNR, α, β , number of corrupted channels, and corrupted time frames increase, indicating that

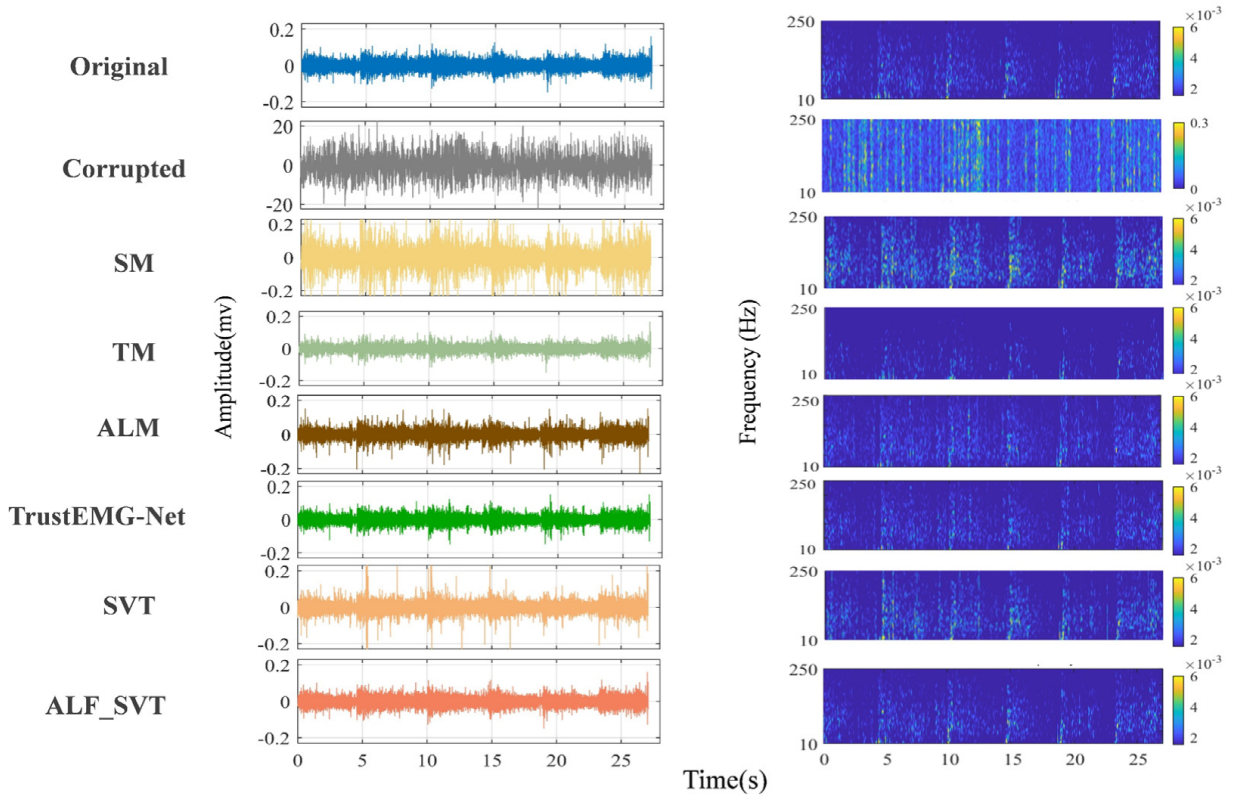


Fig. 8. Time–frequency analysis of restoration signal for different methods.

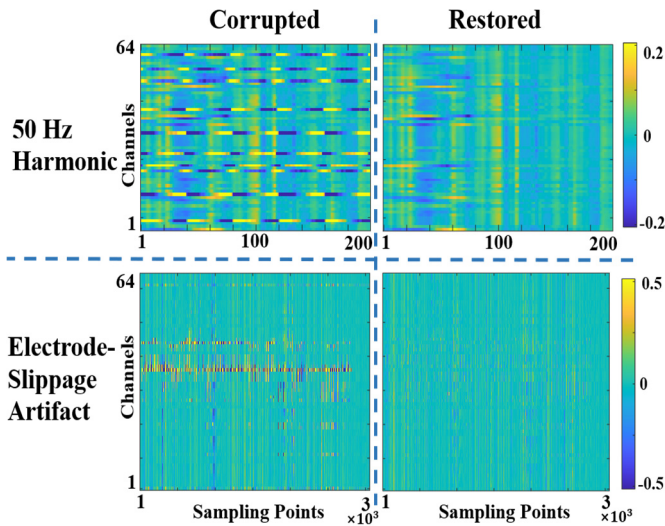


Fig. 9. Restoration results of ALF_SVT under realistic interference conditions.

it is less sensitive to both the patterns and the amplitude of corrupted data, highlighting its ability to adapt to diverse fault scenarios. Although the recovery limits of the method decrease under complete corruption ($\alpha = 100\%$)—as matrix completion relies on partially observed entries to infer missing values based on low-rank structure, and the complete loss of all data in a channel prevents accurate estimation of its

contribution [28]—the sliding window repair strategy (with half window-length overlap, using the second half of the previous window as the first half of the next) enables partial reconstruction by leveraging valid data from adjacent segments, thereby preventing matrix completion failure even when entire channels are missing, thus avoiding $\alpha > 50\%$ corruption within any single window, as shown in Fig. 6(b).

Furthermore, to illustrate the detailed differences in restoration behavior among the six methods (TM, SM, TrustEMG-Net, ALM, SVT, and ALF_SVT), Fig. 8 presents reconstruction results in both time and time–frequency domains. The TM-restored signal shows reduced amplitude and over-smoothing in the time domain. In the time–frequency domain, TM also concentrates spectral energy in low-frequency bands, blurring transient details. SM shows large fluctuations and unstable patterns in both time and frequency domains due to its limited capacity to model spatial correlations across channels. TrustEMG achieves stable amplitude recovery overall; however, its restoration remains limited due to the lack of spatial modeling under large-scale corruption. ALM and SVT perform well in some cases, but their effectiveness decreases in certain windows, possibly due to the concentration of corrupted rows and columns in the 2-D matrix. For example, in a 64-channel setup, if the random five corrupted channels are concentrated within channels 1–5, it leads to excessive sparsity in that region, distorting the singular value spectrum and undermining the low-rank structure, which negatively impacts the reconstruction quality [25]. Moreover, ALF_SVT achieves

more complete restoration in both time and time–frequency domains, closely preserving the signal morphology and spectral structure.

To verify the performance of this repair strategy, in terms of recognition accuracy, as shown in Table II, ALF_SVT achieves 95.5% accuracy when restoring corrupted signals whose accuracy was only 47.7%. The performance is only 1.4% lower than the 96.9% accuracy obtained using clean, uncorrupted signals. Moreover, ALF_SVT significantly outperforms other restoration methods, yielding relative gains of 9.1%, 6.3%, 5.2%, 14.4%, and 12.7% over TM, SM, TrustEMG-Net, ALM, and SVT, respectively. Such improvements in signal restoration contribute to improved reliability and robustness of EMG-based measurement systems.

The limitation of this study is that the performance of the proposed ALF_SVT framework was primarily evaluated under controlled experimental conditions, and under complex mixed-noise conditions (approximately 49% of data points corrupted), the method can still reduce the error rate to 0.28; however, when the corruption proportion increases further, the observed data becomes too sparse, limiting recovery quality. Future work will focus on evaluating the robustness of ALF_SVT with higher-density electrode arrays in real application scenarios for EMG signal measurement, and its potential integration with deep learning methods for further performance enhancement [39], [40], [41]. In addition, we will extend the application of the proposed ALF_SVT framework to various types of multichannel physiological signals, such as electroencephalography (EEG), functional near-infrared spectroscopy (fNIRS), and magnetoencephalography (MEG), to enhance robust detection across diverse real-world conditions.

V. CONCLUSION

This article proposes an adaptive 2-D matrix domain HD-sEMG signal restoration strategy based on low-rank matrix completion, targeting the challenges of complex and high-amplitude noise as well as signal loss commonly encountered in multichannel flexible electrode applications. By leveraging the temporal stability and spatial correlation, the method effectively enhances signal quality under mixed fault conditions and outperforms traditional restoration approaches. Experimental results confirm that the proposed method achieves accurate signal restoration under various types of loss or corruption. The outcomes of this study provide reliable inputs for downstream HD-sEMG applications, supporting high-quality and robust signal measurement.

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